G4-nono cross-domain RECOMBINE pre-registration — frozen snapshot

docs/osf-prereg-g4-nono-cross-domain-2026-10-23.md at commit 14c3d5f86443 (file SHA-256 9c514909…57e9b28a)

G4-nono cross-domain RECOMBINE pre-registration

Split: 2026-10-28, moved from branch `docs/osf-prereg-2026-10` (PR #41) so it can be filed without waiting for E1; only change: the three §9 smoke-record pointers now name `docs/milestones/g4-nono-smoke-records-2026-10/` (archived from macM1 by PI decision, SHA-256 in its `SHA256SUMS`). **Status:** DRAFT — not yet filed on OSF (to be filed by the PI). **Date:** 2026-10-23 (revision 4: PI decisions of 2026-10-23 on N and ε applied; ε reverted to 0.01 and N = 108 tier added; revision 5, 2026-10-24: pre-filing code status, no protocol change; revision 6, 2026-10-24: PI decision removes the N = 30 line of the N-selection rule, so N is 95 or 108; revision 7, 2026-10-24: PI amendment after the first smoke — epochs 3 → 10, acc20 and DOA descriptive only, and P_max runs recombine_full so that P_equ and P_max differ on the CNN; revision 9, 2026-10-25: third-smoke N selection re-checked with the current aggregator, no protocol change) **Parent OSF:** 10.17605/OSF.IO/Q6JYN **Sister pre-regs (locked, not modified):** `docs/osf-prereg-g4-sexto-pilot.md` (H6-A, CIFAR-100), `docs/osf-prereg-g4-septimo-pilot.md` (H6-B / H6-C, Tiny-ImageNet). Their power sentences carry an arithmetic slip; see `docs/osf-prereg-erratum-g4-power-2026-10-23.md` (the locked files are not edited). **c_version at drafting:** C-v0.25.0+PARTIAL **Substrate:** MLX small CNN (`G4SmallCNN`, `experiments.g4_quinto_test.small_cnn`) with `n_classes = 20`, `latent_dim = 64`: one shared 20-way head over coarse labels (class-incremental, no per-task label remap). **Benchmark:** two-domain stream, CIFAR-10 (grayscale) then Quick, Draw! (§5; PI decision 2026-10-24). **Design:** matched-block (seed = block), paired analysis (§2, §6). **Compute:** N = 95 seeds × 9 cells per seed (8 confirmatory + 1 exploratory un-mixed control) = 855 cells, or N = 108 (972 cells) under the pre-stated N-selection rule (§3); 760 / 864 confirmatory cells. Per-cell time at 10 epochs on macM1 under N10-C load: about 41 s on the new corpus (≈ 10 h for 855 cells; third smoke, §9). **Domain corpus:** CIFAR-10 (domain A, grayscale) + Quick, Draw! (domain B), pinned (§5). **Power script:** `scripts/prereg_power_2026_10.py` (§3). **Lock commit:** *(filled by introducing commit hash at filing)* **Lock timestamp:** *(filled at filing, pre-driver-run)*

§1 Background

The G4 escalation ladder (G4-quater → G4-septimo) confirmed the "RECOMBINE-empty" finding (here and below, "confirmed" and "refuted" are the locked decision labels; those non-rejections did not establish equivalence) across {Split-FMNIST, Split-CIFAR-10, Split-CIFAR-100, Split-Tiny-ImageNet} × {3-layer MLP, 5-layer MLP, small CNN, medium CNN} (H6-C CONFIRMED, G4-septimo, Welch p = 0.9247, g = −0.0246). DR-4's prediction "richer ops yield richer consolidation" is recorded as empirically refuted across that ladder (`docs/proofs/dr4-profile-inclusion.md`, DR-4 evidence v0.6). Lemma DR-4.L itself is not refuted.

Every ladder stream is **single-domain** class-incremental, and the scored outcome is retention of earlier tasks. The ladder RECOMBINE strategies (`mog`, `ae`) sample per-class synthetic latents; they never mix classes from different domains.

Zahn, Evans & Eagleman (2026), *Discovery by Dreaming: Cross-Domain Recombination in Artificial Memory* (arXiv:2607.16256, **version 2, 21 July 2026**; v1 was 28 June 2026), report that recombinatory replay **across** domains helps both a LoRA pipeline and a symbolic engine, while **within-domain** rehearsal does not. G4-nono is framed as a **vision generality test** of that claim (class-incremental Split-CIFAR-100-coarse, small CNN), not as a

new hypothesis: the cross- versus within-domain idea is published for LLM and symbolic systems, the vision test is not.

What v2 retains, and what it does not. The v2 full text retracts several numbers that the v1 abstract carried, and the abstract text still shown on the arXiv listing page predates that revision. Only the following v2 evidence is relied on here, as read in the v2 full text (table "Provenance audit", §3.3-§3.5, §5.1, §5.3):

- Neural arm, Llama-3.1-8B, LoRA rank 256, matched conditions: +5.64 ± 2.31 pp (5 of 5 seeds positive, paired t $p = 0.0055$, bootstrap 95 % CI [+3.56, +6.81] pp). The authors state the gain is averaged over a task mix only partly cross-domain; on subtasks requiring cross-domain transfer (e.g. unseen GSM8K reasoning) the gain is reported as +14.5 pp.
- At rank 128 the neural arm is null across three base models, which the authors attribute to capacity under-saturation; the within-domain null they cite (-1.8 ± 4.4 pp, $n = 3 \times 150$, $p > 0.30$) is a rank-128 matched-condition result. At 70B/72B scale the results are inconclusive within a ±5 pp floor.
- Symbolic arm: replay-surfaced bridges lie farther apart in embedding space than within-domain pairs (Spearman $\rho = 0.54$, $n = 32$, bootstrap 95 % CI [0.24, 0.77]) and survive a matched scramble control (5 of 5 true domain-pairs versus 0-1 of 5 re-paired; Fisher exact $p \approx 0.05$). The authors describe the external OpenAlex placement as consistent with, not proof of, the mechanism.
- Weights, not prompt: prepending cross-domain material in-context to a ~671B model gave −9.2 pp (5 seeds). This is a different control from this study's.
- **Retracted in v2 and not used here:** the LLM-judged symbolic accuracy headline (85.7 %, +21 pp over baseline; the authors state judged connection rates are ceilinged near 100 % in all conditions), the single-seed +26.8 % / +19.5 % neural gains, the 57.5 → 74.4 % sequential figure, and the magnitude of the +7.2 % component-synergy figure.

G4-nono asks a narrower, operational question: on a two-domain CIFAR-100-coarse stream, does mixing buffer latents **across** the two domains improve the network's ability to **discriminate a class of one domain from a class of the other** (CDPA, §6), beyond no recombination and beyond matched within-domain mixing?

Scope of the claim. CDPA is a two-class discrimination score computed on one corpus. It does **not** measure cross-domain knowledge transfer, analogy or "discovery" in the sense of Zahn et al. A confirmed H12 supports the statement "cross-domain latent mixing improves cross-domain class discrimination on this stream", and only that statement can be used to re-scope the DR-4 evidence record. No transfer endpoint is pre-registered here.

Domain-shortcut confound (Zhang et al., arXiv:2607.22994). Zhang, Fan, Liang et al., *Breaking the Synthetic-Real Domain Shortcut for Training-Free Generative Replay-based Class Incremental Learning*, show that mixing synthetic old-class data with real new-class data lets a model rely on domain-discriminative features instead of class cues. CDPA is an A-versus-B discrimination score, so a `cross` gain could reflect a domain shortcut or head recalibration (the `fc2` soft-target step is a logit-rebalancing step between domains, which may counteract recency bias toward domain B) rather than recombination. A confirmed H12-A does not exclude these mechanisms; the domain-only probe of §6 is the check.

The LLM-tier variant (LoRA math → code) is out of scope: in the CUDA E1 harness (`scripts/pilot_cycle3_cuda.py`, `_apply_dream_ops`) `recombine` is a weight-neutral no-op.

§2 Hypotheses (confirmatory)

Strategies: `cross` (cross-domain latent mixing), `within` (matched within-domain mixing), `none` (RECOMBINE skipped — the ladder's H4-C / H5-C / H6-A placebo). All confirmatory contrasts are in the `P_max` arm.

Pairing. Within a seed, every cell uses the same initial weights (`G4SmallCNN(seed)`), the same per-task data order (`np.random.default_rng(seed)` inside `train_task`), the same buffer fill draws (`default_rng(seed + 5000)`) and the same dream-episode seeds. `cross`, `within` and `none` are therefore **matched**, not independently randomised, and are bit-identical up to the first dream episode where they differ (§5). The unit of analysis is the seed; every contrast is analysed on the per-seed difference `D_s = Y_s(strategy 1) − Y_s(strategy 2)`.

Family. Three hypotheses, Bonferroni $\alpha = 0.05 / 3 =$ **1/60 ≈ 0.0167** per hypothesis. For each contrast let `L` and `U` be the one-sided $(1 - \alpha)$ lower and upper confidence bounds for the mean of `D` (paired t, df = n − 1), i.e. the limits of the two-sided $(1 - 2\alpha) = 96.67$ % interval. Effect sizes:

- **g_av** (primary effect size, comparable to the ladder's between-group g): `mean(D) / sqrt((s₁² + s₂²) / 2) × J`, with `J = 1 − 3 / (4(n − 1) − 1)` (Lakens 2013 average-SD form).
- **g_z** (reported): `mean(D) / sd(D) × J`.

Degenerate case. If every retained D is identical (`sd(D) = 0`), the t statistic is undefined; then `L = U = mean(D)` and the state follows from these bounds. An undefined g_z or g_av (zero denominator) is reported as undefined and does **not** meet the `g_av ≥ 0.5` floor.

Margin. ε = **0.01** (one percentage point of pairwise accuracy, raw units; PI decision of 2026-10-23, which first halved it to 0.005 and then reverted it to 0.01 after seeing the power numbers of §3). Below ε a change in CDPA or WDPA is declared practically negligible for the DR-4 record. It is a substantive choice, not derived from data; power against it is tabulated in §3.

- **H12-A (cross-domain gain)** — `CDPA(cross) − CDPA(none) > 0`. Paired one-sided t-test (`scipy.stats.ttest_rel(..., alternative="greater")`) at α = 1/60.
- **H12-B (cross-domain specificity)** — `CDPA(cross) − CDPA(within) > 0`. Same test.
- **H12-C (new within-domain equivalence check, bounded)** — tests WDPA under latent mixing, not the ladder's retention endpoint — `|WDPA(within) − WDPA(none)| < ε`. Paired TOST at α = 1/60 per side: equivalence iff `L > −ε` and `U < ε`.

Per-hypothesis states (evaluated in the order listed; the first that applies is the state; the list is exhaustive):

For H12-A and H12-B:

State	Rule
CONFIRMED	`L > 0`, observed `g_av ≥ 0.5` (the G4-octavo H7-A / G6 H9-A floor) **and** observed `mean(D) ≥ ε`
POSITIVE-NEGLIGIBLE	`L > 0` and `U < ε` (positive but bounded below the practical margin)
DETECTED-BELOW-FLOOR	`L > 0` and neither of the above
BOUNDED-NO-GAIN	`U < ε` (no gain of ε or more; includes negative estimates)
INCONCLUSIVE	none of the above

What CONFIRMED means. `L > 0` establishes a positive population difference at level α; the observed `g_av ≥ 0.5` and `mean(D) ≥ ε` are magnitude filters on the **point estimate** (standardised and raw), so a tiny but very precise difference cannot be CONFIRMED. CONFIRMED is therefore read as "a positive effect whose estimate reaches the floor", never as "the population effect is at least 0.5". The same applies to every observed-floor rule in this document.

If `U < 0`, the state is BOUNDED-NO-GAIN and the negative direction is reported as an **exploratory** harm signal. No harm conclusion is confirmatory in this family.

For H12-C:

State	Rule	Status
EQUIVALENT	`L > −ε` and `U < ε`	confirmatory
MEANINGFULLY-DIFFERENT	`L > ε` or `U < −ε`	exploratory
INCONCLUSIVE	none of the above (includes a difference from 0 whose interval still overlaps ±ε)	—

Error accounting. Each hypothesis spends α once. For H12-A/B the confirmatory conclusions are CONFIRMED (false only if the true difference μ ≤ 0), BOUNDED-NO-GAIN (false only if μ ≥ ε) and POSITIVE-NEGLIGIBLE (their conjunction). Because ε > 0, at most one of them can be false for any μ, so the probability of a false confirmatory conclusion is ≤ α per hypothesis. For H12-C only EQUIVALENT is confirmatory (TOST, α); MEANINGFULLY-DIFFERENT is exploratory and spends no α. The family bound is therefore 3α = 0.05.

A non-significant test never establishes a null in this document. Only BOUNDED-NO-GAIN and EQUIVALENT, which rest on confidence bounds against ε, support "no effect of practical size". A difference from zero that is smaller than ε does not count as non-replication.

Secondary contrasts (reported, outside the family, same state rules at nominal α = 0.05): `CDPA(P_max, cross)` vs `CDPA(P_equ, cross)`, `CDPA(P_max, cross)` vs `CDPA(P_min)`, `CDPA(P_max, within)` vs `CDPA(P_max, none)` (needed before any "generic synthetic-SGD" reading), and the same contrasts on retention. Final 20-way accuracy (acc20) and the domain-only probe (DOA) are **descriptive only** (PI decision 2026-10-24): mean, SD and a two-sided 95 % paired interval, no state (see §6 for why).

§3 Power analysis

All numbers are reproduced by `scripts/prereg_power_2026_10.py` (exact noncentral t for test-only MDEs; Monte Carlo, 20 000 or 200 000 draws, for complete rules).

Test-only MDE at 80 % power, one-sided α = 1/60. The paired test needs `d_z = 0.308` (g_z ≈ 0.306) at n = 95 and `d_z = 0.289` (g_z ≈ 0.287) at n = 108. Converted to the g_av scale with `d_av = d_z × sqrt(2(1 − r))`, where r is the within-seed correlation between the two strategies' outcomes:

n (d_av MDE)	r = 0	r = 0.3	r = 0.5	r = 0.7	r = 0.9
95	0.436	0.365	0.308	0.239	0.138
108	0.408	0.342	0.289	0.224	0.129

For reference, an independent-groups design would need d = 0.434 at n = 95 per group and d = 0.406 at n = 108.

Complete confirmation rule (paired t rejects **and** `g_av ≥ 0.5`), power by true d_av and r:

n = 95	d_av = 0.5	0.6	0.8	1.0
r = 0	0.49	0.74	0.98	1.00
r = 0.3	0.49	0.78	0.99	1.00
r = 0.5	0.49	0.82	1.00	1.00
r = 0.7	0.49	0.87	1.00	1.00
r = 0.9	0.48	0.95	1.00	1.00

n = 108	d_av = 0.5	0.6	0.8	1.0
r = 0	0.49	0.76	0.98	1.00
r = 0.3	0.50	0.79	0.99	1.00
r = 0.5	0.49	0.83	1.00	1.00
r = 0.7	0.49	0.89	1.00	1.00
r = 0.9	0.49	0.97	1.00	1.00

A true effect exactly at the floor (d_av = 0.5) is confirmed at most about half the time at any n; this is a property of an observed- effect floor, stated here so it is not read as a design flaw later.

The raw floor `mean(D) ≥ ε` (ε = 0.01) only bites when arm SDs are small. With r = 0.5, n = 95 and power shown by true d_av:

arm SD	d_av = 0.5	0.6	0.8	1.0
0.01	0.00	0.00	0.03	0.50
0.02	0.45	0.80	1.00	1.00
0.04	0.49	0.82	1.00	1.00

When the seed SD of CDPA is ≈ 0.01, a standardised effect of 0.5-0.8 is only 0.005-0.008 accuracy points, i.e. below ε, and is by design not confirmable. From an arm SD of 0.02 upwards the raw floor barely changes the complete-rule power.

"No practical effect" rules at ε = 0.01 (true difference 0; all powers in this section assume the full planned N is retained): H12-C EQUIVALENT (paired TOST, Monte Carlo) and, for H12-A/B, two events. $U < \varepsilon$ (exact noncentral t) is the union of BOUNDED-NO-GAIN and POSITIVE-NEGLIGIBLE under the first-match order of §2; the exact BOUNDED-NO-GAIN state is $L \le 0 \wedge U < \varepsilon$ (Monte Carlo), which at a true difference of 0 is capped near $1 - \alpha = 0.983$ because $L > 0$ then sends the outcome to POSITIVE-NEGLIGIBLE. Power by SD of the paired difference SD_D:

rule, N	SD_D = 0.0075	0.01	0.015	0.02	0.025	0.03	0.04
EQUIVALENT, 95	1.00	1.00	1.00	0.99	0.92	0.72	0.22
EQUIVALENT, 108	1.00	1.00	1.00	1.00	0.95	0.81	0.34
U < ε (BNG ∪ PN), 95	1.00	1.00	1.00	1.00	0.96	0.86	0.61
U < ε (BNG ∪ PN), 108	1.00	1.00	1.00	1.00	0.98	0.90	0.67
BOUNDED-NO-GAIN (exact), 95	0.98	0.98	0.98	0.98	0.96	0.86	0.61
BOUNDED-NO-GAIN (exact), 108	0.98	0.98	0.98	0.98	0.97	0.90	0.67

Largest SD_D with 80 % power (EQUIVALENT / $U < \varepsilon$): 0.0282 / 0.0324 at N = 95, 0.0301 / 0.0346 at N = 108. Seeds needed for 80 %: 29 / 23 at SD_D = 0.015, 49 / 38 at 0.02, 75 / 58 at 0.025, 108 / 82 at 0.03, 189 / 144 at 0.04.

N-selection rule (PI decisions of 2026-10-23 and 2026-10-24). The pre-filing smoke (§9, seeds 900-902, outside the confirmatory list, never used for inference) computes from its three seeds SD_D, the sd (ddof = 1) of the three per-seed differences `WDPA(within) − WDPA(none)`. N is then fixed by the first matching line:

1. SD_D ≥ 0.015 → **N = 108** (seeds 0..107);
2. otherwise → **N = 95** (seeds 0..94).

The rule is applied once, before any confirmatory cell runs; its inputs and outcome are written into the filed version, and N is fixed from then on. **Smoke validity.** N is selected only from a complete smoke: all 24 cells (8 per seed) present, with status `ok` and finite CDPA / WDPA. The registered abort conditions of §4 apply to the smoke: a run_id collision, or a manipulation-check failure on 2 or 3 of the 3 smoke seeds, refuses the selection (the cause is investigated and the amendment path of §4 applies). A single seed failing the manipulation check is dropped and SD_D is computed from the other two (ddof = 1); this is reported. All 24 cells are validated first, including those of a dropped seed. Any other missing or unusable cell (no record, non-finite metric, excluded cell) also refuses the selection: the cause is fixed and the full smoke is rerun in a fresh output directory and registry, and both smoke runs are reported; a driver change made for this is part of the lock commit. With three seeds SD_D is very imprecise: the rule is a pre-stated deterministic trigger, not an estimate of the population SD_D. No ladder milestone stores per-seed paired CDPA or WDPA differences (the metrics are new), so there is no prior estimate; the planning value for SD_D is the threshold, 0.015. The smoke also reports $\hat{r}$, the Pearson correlation across the smoke seeds between the two cells of each confirmatory contrast (CDPA `cross`/`none`, CDPA `cross`/`within`, WDPA `within`/`none`; an undefined $\hat{r}$ is reported as such). $\hat{r}$ is descriptive only: it no longer selects N.

Why the former N = 30 line was removed (PI decision, 2026-10-24). The revision-4 rule had a third line (N = 30 when SD_D < 0.015 and every $\hat{r} \ge 0.5$). A three-seed sd often falls below the threshold when the true SD_D is above it: with normal D, $2\,SD_D^2 / SD_D^2$ follows χ^2_2, so $P(SD_D \ge 0.015) = \exp(-(0.015 / SD_D)^2)$, which is only 0.37 at SD_D = 0.015 and 0.57 at 0.02. When the $\hat{r}$ line did not trigger, the old rule therefore stayed at N = 30 too often, and its unconditional EQUIVALENT power was 0.89 / 0.74 / 0.69 / 0.63 at SD_D = 0.015 / 0.02 / 0.025 / 0.03.

Unconditional power of the registered rule. Method: the selection is driven by σ_sel, the SD of `WDPA(within) − WDPA(none)`; $P(N = 108) = \exp(-(0.015 / \sigma_sel)^2)$ from the χ^2_2 law above. The smoke seeds are disjoint from the confirmatory seeds, so a rule's power is P(N = 95) × power(95) + P(N = 108) × power(108), with power(n) evaluated at the SD of that contrast, σ_con. True difference 0, normal D, ε = 0.01, α = 1/60, full planned N retained (`scripts/prereg_power_2026_10.py`, section 5b).

H12-C. Its contrast is the selector difference, so σ_con = σ_sel = SD_D:

SD_D	0.0075	0.01	0.015	0.02	0.025	0.03	0.04
P(N = 108)	0.02	0.11	0.37	0.57	0.70	0.78	0.87
EQUIVALENT	1.00	1.00	1.00	0.995	0.94	0.79	0.33

H12-A/B. The CDPA contrast SD σ_con is a different quantity from σ_sel, so power is given for separate scenarios (rows σ_sel, columns σ_con), for the exact BOUNDED-NO-GAIN state:

σ_sel σ_con	0.01	0.015	0.02	0.025	0.03	0.04
0.0075	0.98	0.98	0.98	0.96	0.86	0.61
0.015	0.98	0.98	0.98	0.96	0.88	0.63
0.03	0.98	0.98	0.98	0.97	0.89	0.66

(U < ε, i.e. BOUNDED-NO-GAIN ∪ POSITIVE-NEGLIGIBLE, is the same except 1.00 in place of 0.98 for σ_con ≤ 0.02.)

What the rule can and cannot establish (stated plainly). With the 95-seed floor, H12-C EQUIVALENT keeps at least 80 % power for SD_D up to 0.025 (0.94) and reaches 0.79 at 0.03; H12-A/B BOUNDED-NO-GAIN keeps at least 80 % for a CDPA contrast SD up to 0.03 (0.86-0.89 depending on σ_sel). At larger SDs the design is under-powered: at 0.04 EQUIVALENT power is 0.33 (189 seeds needed) and BOUNDED-NO-GAIN 0.61-0.66 (144 needed). The complete H12-A/B confirmation rule has about 0.5 power at a true d_av exactly at the 0.5 floor at either N (table above); this is a property of the observed-effect floor, not of N. Retained n below the planned N (exclusions, §4) lowers every power in this section.

Under-power makes INCONCLUSIVE more likely; it does not change the error rates of §2.

§4 Exclusions, replacement and retained-N inference

- **Learning manipulation check (seed level).** Right after task 1, before any dream episode, accuracy on the 1000 task-1 test images is scored **restricted to the two task-1 logits** (2-way). The seed is excluded from every contrast if this accuracy is below **0.60**. Rationale: after task 1 the other 18 output units are untrained, so the meaningful chance level is the 2-way 0.50, not the 20-way 0.05 inherited from the ladder's "2 × chance" rule. With 1000 test images the binomial SD at chance is ≈ 0.016, so 0.60 is ≈ 6 SD above chance; it checks that learning happened, nothing more. This check precedes every treatment difference, so it is identical for all eight cells of a seed and cannot bias a contrast.
- **Cell-level exclusions.** A cell is excluded if CDPA, WDPA or final accuracy is non-finite, or if a BLOCKING invariant (I1, I2, S1, S2, S3, K1, K4) is violated (`docs/osf-preregistration-draft.md` §4). A contrast uses the seeds for which **both** of its cells are retained (pairwise deletion). The estimand is therefore the mean paired difference over seeds whose two cells both complete, and the error rates of §2 are conditional on that set; the guard below limits, but does not remove, selection by treatment-dependent failures. A cell whose run raises (for example a BLOCKING invariant raised by the dream runtime) writes no outcome record; the aggregator counts the missing record as a cell-level exclusion (reason `missing_record`) and the driver log is preserved.
- **No replacement.** The seed list is fixed (§5). Excluded seeds are not replaced. Inference uses the retained n; the achieved n and every excluded seed with its reason are reported.
- **Differential-exclusion guard.** If a confirmatory contrast loses more than 9 of its 95 seeds (more than 10 of 108) to **cell-level** exclusions, that hypothesis's state is still reported but is labelled exploratory: cell-level exclusions can depend on the treatment. The §7 row is still computed from the raw states, but its reading is **exploratory** whenever a hypothesis that the row's first-match condition reads is guarded (row 1 reads H12-C; rows 2-5 read H12-A, H12-B and H12-C; rows 6-8 read H12-A and H12-C). An exploratory row is reported with its number and never supports a re-scoping (row 2), an addendum (rows 3-5) or an extension (row 6). The aggregator writes `row_confirmatory` and `row_guarded_by`.

- **Worst-case imputation sensitivity (PI decision, 2026-10-24; reported only).** For each H12 contrast, every cell lost at cell level among the retained seeds (excluded cell or missing record; seed-level manipulation exclusions are not imputed) is imputed against the treatment: the treatment cell (`P_max/cross` for H12-A/B, `P_max/within` for H12-C) takes the lowest value observed for that cell over the retained seeds, and the comparator cell the highest. The §2 states and the §7 row are recomputed on the imputed data and reported next to the primary ones. The observed-extrema imputation is a directional sensitivity analysis, not an absolute worst-case bound (the true lost values could lie outside the observed range, and for the H12-C equivalence claim the unfavourable direction is not unique). This analysis never changes a decided state, the row or its confirmatory status; a disagreement with the primary reading is reported in the milestone and in Paper 2.
- **Abort.** If more than 50 % of seeds fail the manipulation check, or a run_id collides with a prior registry entry, the run is aborted and amended (raise epochs per G4-sexto §9.1). On abort, **all artifacts are preserved**: every completed cell JSON, the registry rows, driver logs and the partial aggregate are written to `docs/milestones/g4-nono-ABORTED-<date>.{json,md}`, labelled ABORTED and non-confirmatory, and the amendment cites that file. No confirmatory milestone is written for an aborted run.

§5 Stream, substrate and driver (executable specification)

- **Seeds.** Confirmatory seeds `0..N − 1`, with N ∈ {95, 108} fixed by the §3 N-selection rule. Smoke seeds `900..902` (never analysed).
- **Domain corpus (PI decisions 2026-10-24; revision 8).** The CIFAR-100 animals / non-animals split is replaced by two corpora with a much stronger shift: **domain A = CIFAR-10** (natural photographs) and **domain B = Quick, Draw!** (hand-drawn doodles). Module `experiments/g4_nono_test/qd_cifar10.py`, build script `scripts/g4_build_qd_cifar10.py`.
  - Domain A: the 10 CIFAR-10 classes, HF `uoft-cs/cifar10` at revision `0b2714987fa478483af9968de7c934580d0` parquet SHA-256 train `8428b53a…d2ae883`, test `841389e6…645ce84de` (the G4-quinto pins). CIFAR-10 states no licence; its page asks for a citation (Krizhevsky 2009), the same exposure as CIFAR-100 before.
  - Domain B: Quick, Draw! `numpy_bitmap` (28 × 28 grayscale, Google Creative Lab, **CC BY 4.0**; attribution: "Quick, Draw! dataset, Google, CC BY 4.0"), categories apple, banana, clock, cup, house, key, ladder, moon, tree, umbrella (semantically disjoint from CIFAR-10; no vehicle or animal category). No official SHA-256 exists: our SHA-256 of each `.npy`, its byte size and the GCS md5 / crc32c are pinned in `QD_PINS` (downloaded 2026-10-24; the md5s match GCS).
  - **Colour shortcut neutralised.** CIFAR-10 images are converted to grayscale (ITU-R BT.601 luma, rounded) and replicated to 3 channels; drawings are zero-padded 28 → 32 (2 px each side) and replicated to 3 channels. Every image of both domains therefore has three equal channels, so the domain cannot be read from colour alone (tested). Other low-level cues remain (binary strokes on a black background vs grey-level photographs); they are part of the stronger shift the PI chose. **Scope of the claim.** On this corpus G4-nono is an operational CDPA test: it can show whether cross-domain mixing changes cross-domain pairwise discrimination, but it cannot establish semantic integration across domains, nor exclude that a gain comes from recalibrating on domain cues.
  - **Subsampling.** Each class is cut to exactly **2 500 train / 500 test** images with `default_rng([20261024, corpus, class])` (CIFAR-10: from its train and test splits; Quick, Draw!: disjoint draws from the full file). The index lists are committed (`experiments/g4_nono_test/qd_cifar10_indices.json`, SHA-256 `f4092ff2…eb11ae7`); the built corpus `qd-cifar10-v1.npz` (60 000 images, SHA-256 `4f7fdf1c…bf381b`) is verified by the loader and rebuilds bit-identically. Raw `.npy` files are deleted after the build (about 1.2 GB downloaded once; 185 MB kept). Per-task image counts equal the CIFAR-100 design (5 000 train / 1 000 test), so per-cell compute is unchanged.
  - **Label map.** The coarse-id space is kept so the stream, metrics and RECOMBINE code are unchanged: CIFAR-10 class k → id `DOMAIN_A[k]` (airplane 0, automobile 1, bird 7, cat 8, deer 11, dog 12, frog 13, horse 14, ship 15, truck 16); Quick, Draw! class k → `DOMAIN_B[k]` (apple 2, banana 3, clock 4, cup 5, house 6, key 9, ladder 10, moon 17, tree 18, umbrella 19).
- **Stream.** 10 tasks of 2 classes, consecutive in id order within each domain: tasks 1-5 = A {0,1}, {7,8}, {11,12}, {13,14}, {15,16} (airplane/automobile, bird/cat, deer/dog, frog/horse, ship/truck); tasks 6-10 = B {2,3}, {4,5}, {6,9}, {10,17}, {18,19} (apple/banana, clock/cup, house/key, ladder/moon, tree/umbrella). Each task uses its 5 000 training and 1 000 test images. Labels are the ids above on the 20-way head. No per-seed reshuffle of

task composition. The previous CIFAR-100 split remains selectable (`--corpus cifar100`) only to reproduce the first two smokes.

- **Training per task.** `train_task(epochs = 10, batch_size = 64, lr = 0.01)`, plain SGD, cross-entropy, no augmentation, input scaling as in `experiments.g4_sexto_test.cifar100_dataset` (G4-sexto defaults). The last partial batch is kept. **Amendment (PI decision 2026-10-24): epochs 3 → 10.** The first smoke (§9) hit the §4 manipulation-check abort at 3 epochs (2 of 3 seeds below 0.60). Short reruns showed genuine under-fitting of the first pair (train and test 2-way accuracy equal, loss near ln 2) and task-1 2-way accuracy 0.615 / 0.639 at 10 epochs on seeds 900 / 901 (0.609 / 0.500 at 3 epochs, 0.601 / 0.613 at 8). Precedent: G4-sexto §9.1 raised epochs (3 → 8) for the same substrate and loader. The change applies to every arm and strategy alike, so it cannot favour one side of any contrast. Driver default `EPOCHS = 10`.
- **Dream episodes (ladder schedule).** One DE after each of tasks 1-9, i.e. before training tasks 2-10, with DE seed `seed + k` for the DE that precedes task k + 1 (k = 1..9). No DE after task 10. Operations per arm as in the G4-sexto driver: `P_min` = REPLAY + DOWNSCALE; `P_equ` and `P_max` = REPLAY + DOWNSCALE + RESTRUCTURE + RECOMBINE; `baseline` = no DE. Constants: replay HP combo C5 (`representative_combo()`: downscale 0.95, replay batch 32, 5 steps, lr 0.01), `RESTRUCTURE_FACTOR = 0.05`, `RECOMBINE_N_SYNTHETIC = 16`, `RECOMBINE_LR = 0.01`.
- **β buffer.** Capacity 320, never evicts in this stream. After each task, 32 training records are pushed, **16 per superclass**, drawn without replacement with `default_rng(seed + 5000)`. Each record stores the image and its 64-d post-ReLU `fc1` latent computed by the network **at push time**. Latents are **never refreshed** (ladder policy); domain-A latents mixed at later DEs were computed by an earlier network. This is part of what is tested.
- **RECOMBINE strategies** (extension of `experiments/g4_quater_test/recombine_strategies.py`; module `experiments/g4_nono_test/recombine_cross.py`, written and reviewed against the checklist below in PR #44). At a DE with recombine seed `seed + k + 30000`, one `default_rng` stream is used in this order:
  1. draw $\lambda_1..\lambda_{16} \sim$ Uniform(0.2, 0.8) (so `cross` and `within` use identical λ values within a seed);
  2. draw the 16 class pairs;
  3. draw one buffer record per class in each pair, uniformly among that class's records.
  - `cross`: pair j = (a, b) with a uniform over domain-A classes present in the buffer and b uniform over domain-B classes present. Synthetic latent `z = λ_j z_a + (1 − λ_j) z_b`, soft target `λ_j e_a + (1 − λ_j) e_b` (20-d). Active only once both domains are in the buffer, i.e. at the DEs before tasks 7-10 (k = 6..9). At k = 1..5, `cross` is a no-op, identical to `none`.
  - `within`: pairs 1-8 are two **distinct** domain-A classes, pairs 9-16 two distinct domain-B classes (drawn without replacement within the pair). Same λ, same target construction, same DEs (k = 6..9), so `cross` and `within` have identical sample counts and SGD steps.
  - `none`: RECOMBINE skipped.
  - **Profile variant (PI decision 2026-10-24).** Framework C §3.1 gives `P_equ` the op recombine_light and `P_max` recombine_full; the op sets otherwise coincide. On the CNN path `P_equ` uses the interpolant above (light). `P_max` (full) draws, after the record draws of the same stream, ε ~ N(0, I) (16 × 64) and uses `max(0, z + σ ⊙ ε)`: a reparameterised Gaussian sample around the interpolant, with σ the per-dimension SD of all buffer latents (a moment-matched encoder) and the `fc1` ReLU as decoder, so the sample stays in the post-ReLU latent space. λ, pairs, records and soft targets are identical across the two variants. Before this change the two profiles differed only in runtime handlers acting on the dummy DE input slice, so every CNN-side update and every metric of `P_equ` and `P_max` was identical; the same holds for the G4-quinto results (retention 0.8754 for both), which are therefore an implementation artefact and are not edited. The primary H12 contrasts are within `P_max`, so their treatment cells now use recombine_full. `P_equ` and `P_max` still coincide in the `none` cells and at the DEs where `cross` / `within` are no-ops (k = 1..5), because RECOMBINE is the only op that differs. `baseline`, `P_min`, every `P_equ` cell and `P_max/none` are bit-identical to the code before the change (regression test on macM1; the golden hashes cover a 1-epoch synthetic fixture, not the default 10-epoch CIFAR outputs). The buffer-SD Gaussian sample is this study's documented CNN operationalisation of recombine_full; framework C §3.1 names the op but does not prescribe this form. The α input and channel 2 of the framework's `P_max` are not modelled on the CNN path.
  - Update for `cross` / `within`: one SGD step (lr 0.01) on the head `fc2` only, mean soft-target cross-entropy `−Σ_c t_c log softmax(fc2(z))_c` over the 16 samples.
- **Cells per seed.** `baseline` and `P_min` do not run RECOMBINE, so they are run once per seed. `P_equ` and

P_max are run with each of the three strategies. 2 + 2 × 3 = **8 confirmatory cells per seed** (760 at N = 95, 864 at N = 108), plus the exploratory P_max/unmixed cell (below): **9 cells per seed**, 855 / 972.

- **Exploratory un-mixed control (PI decision 2026-10-24; reported only).** Cell P_max/unmixed: at each DE it makes exactly the cross draws (same stream, λ, pairs and records) but replays the two records of each pair **unmixed**, with hard one-hot labels (32 latents, one fc2 step at lr 0.01, same mean cross-entropy), and with P_max's recombine_full noise drawn for the 32 latents. It separates recombination from logit rebalancing. Contrasts CDPA(P_max, cross) − CDPA(P_max, unmixed) and CDPA(P_max, unmixed) − CDPA(P_max, none) are reported with the §2 state rules at nominal α = 0.05 in an exploratory_arms block. They are never confirmatory, are outside the family, the guard, the §7 row and the completeness check of the 8 registered cells, and never change a decided state. Exploratory records never enter the abort conditions, the seed-level exclusions or the smoke N selection; their own aborted records and manipulation flags are reported in the exploratory block.
- **Driver / aggregator.** experiments/g4_nono_test/run_cross_domain.py (one cell per call, PR #44) and experiments/g4_nono_test/aggregator.py (PR #41). The aggregator implements §2, §4 and §7 exactly; its bounds come from the paired helper kiki_oniric.eval.statistics.paired_tost (paired SE, df = n − 1), whose one-sided bound L > 0 is the same decision as scipy.stats.ttest_rel(..., alternative="greater") at α. Its --select-n mode applies the §3 N-selection rule to the smoke records.
- **Data pin.** qd-cifar10-v1.npz SHA-256 4f7fdf1c…bf381b, built from the pinned CIFAR-10 parquet files and Quick, Draw! .npy files (§5 domain corpus); the driver's default is --corpus qd-cifar10.

Implementation fidelity checklist (all items ticked in the filing commit; each has a unit test under tests/):

1. recombine_cross.py reuses the SyntheticLatentBatch interface but returns soft targets; the ladder mog / ae code paths are untouched.
2. λ is drawn before classes and records (shared λ across cross and within for a seed) — test: same seed ⇒ same λ vector.
3. within never pairs a class with itself; cross never pairs two classes of one domain — test over 10^4 draws.
4. cross and within are no-ops at k ≤ 5 — test: weights after the DE before task 6 are bit-identical to none.
5. The corpus file matches its pinned SHA-256 and holds 2 500 / 500 images per class with three equal channels, and the index file matches its pin — test_checklist_item5_real_corpus (passes on macM1, where the corpus is built; skipped elsewhere). (The former item 5, CIFAR-100 coarse-label decode, applies only to the superseded split.)
6. Buffer push is 16 + 16 per task, latents frozen at push — test.
7. CDPA / WDPA tie rule (§6) — test with constructed equal logits.
8. The aggregator reproduces a hand-computed paired example (L, U, g_av, g_z, state) — test.

Tests per item (status 2026-10-24): 1 test_ladder_module_untouched, test_shapes_and_soft_targets; 2 test_same_seed_same_lambda_across_strategies; 3 test_distinct_class_and_domain_constraints_1e4_draws; 4 test_cross_and_within_noop_before_task6; 5 test_pinned_bin_coarse_matches_parquet_first_1000 (skipped unless both pinned files are present, so it must be run on the filing host; see §9); 6 test_buffer_push_16_plus_16_latents_frozen; 7 test_tie_on_a_single_pair_counts_half_per_image, test_all_equal_logits_give_exactly_half; 8 test_paired_summary_hand_example (tests/unit/test_g4_nono_{recombine_cross,driver,loader_coarse,metrics,aggregator}.py).

§6 Outcome metrics

All metrics are computed after task 10 training (end of stream).

- **CDPA (primary).** For every pair (a, b) with a ∈ A and b ∈ B (100 pairs): binary accuracy over the 1000 test images of a and b, predicting the class with the larger of the two logits {a, b}. An exact logit tie counts 0.5. CDPA is the unweighted mean over the 100 pairs.
- **WDPA (H12-C).** Same computation over the 45 + 45 within-domain pairs; unweighted mean over the 90 pairs.
- **Domain-only probe, DOA (descriptive only, PI decision 2026-10-24; experiments.g4_nono_test.metrics.compute_doa, cell field doa).** For each of the 20000 test images of the 20 classes (1000 per class), predict the *domain* by comparing the maximum logit over the 10 domain-A classes with the maximum over the 10 domain-B classes (tie counts 0.5); DOA is the accuracy against the true domain. It uses no within-

pair class information. The paired contrasts `cross − none` and `cross − within` on DOA are reported descriptively (mean, SD, two-sided 95 % paired interval, no state). With one shared class-incremental head the network predicts almost only the last task's classes (domain B), so DOA is pinned near 0.5 by construction (exactly 0.5 in every cell of the first smoke). Reading rule (descriptive): if the `cross` gain in DOA is of the same size as the gain in CDPA, the CDPA gain cannot be separated from a domain-shortcut or head-recalibration mechanism (Zhang et al., arXiv:2607.22994) and H12-A / H12-B are reported with that qualification. Nothing in §2 or §7 changes.

- **Retention (secondary, ladder-comparable).** Accuracy on domain-A test images after task 10, restricted to the 10 domain-A logits.
- **Final 20-way accuracy, acc20 (descriptive only, PI decision 2026-10-24).** Reported like DOA, no state. With the collapse onto the last task, acc20 is at most 0.1 × the last pair's 2-way accuracy (0.05 in every cell of the first smoke), so it cannot carry a contrast.
- Effect sizes: g_av and g_z as defined in §2 (paired). The ladder's independent-cell Hedges' g is not used for confirmatory contrasts.

§7 DualVer outcome rules

Rows are evaluated top to bottom; the **first matching row** applies. The table is exhaustive: every combination of states reaches a row.

#	Condition	Reading	EC	FC
0	Abort condition of §4	Abort and amend; artifacts preserved as ABORTED. No reading.	n/a	n/a
1	H12-C = MEANINGFULLY-DIFFERENT	Exploratory signal that within-domain mixing changes WDPA by more than ε on this stream (the ladder's within-domain null may not carry over). H12-A/B states are reported but read as exploratory until this is explained.	stays PARTIAL	stays
2	H12-A = CONFIRMED ∧ H12-B = CONFIRMED ∧ H12-C = EQUIVALENT	**DR-4 refutation re-scoped (operationally).** Cross-domain latent mixing gives a positive CDPA difference over no mixing and over within-domain mixing (point estimates at or above the floor), while within-domain mixing changes WDPA by less than ε. DR-4 evidence gets a G4-nono addendum restating the refutation as "refuted for within-domain retention; not refuted for cross-domain discrimination on one CIFAR stream".	stays PARTIAL	stays
3	H12-A = CONFIRMED ∧ H12-B = CONFIRMED ∧ H12-C = INCONCLUSIVE	Cross-domain-specific CDPA gain, but the within-domain null is not bounded. Addendum records the gain; the re-scoping of row 2 is **not** made.	stays PARTIAL	stays
4	H12-A = CONFIRMED ∧ H12-B ∈ {BOUNDED-NO-GAIN, POSITIVE-NEGLIGIBLE}	Gain over `none`; the cross-domain-**specific** advantage over `within` is bounded below ε. No cross-domain specificity is claimed; the DR-4 refutation stands for this endpoint. Whether the gain is generic synthetic SGD is read only from the secondary `within` vs `none` CDPA contrast, not concluded here.	stays PARTIAL	stays
5	H12-A = CONFIRMED ∧ H12-B ∈ {DETECTED-BELOW-FLOOR, INCONCLUSIVE}	Gain over `none`; specificity unresolved. No re-scoping.	stays PARTIAL	stays
6	H12-A ∈ {BOUNDED-NO-GAIN, POSITIVE-NEGLIGIBLE}	No CDPA gain of ε or more from cross-domain mixing. The DR-4 refutation is **extended** to this two-domain stream for the CDPA endpoint.	stays PARTIAL	stays

#	Condition	Reading	EC	FC
7	H12-A = DETECTED-BELOW-FLOOR	A positive CDPA difference whose estimate misses a floor and is not bounded below ε. Reported; no re-scoping, no extension.	stays PARTIAL	stays
8	H12-A = INCONCLUSIVE	Inconclusive at this N: neither a gain ≥ floor nor a bound below ε. DR-4 record unchanged; no extension is claimed.	stays PARTIAL	stays

No row promotes EC to STABLE. A single two-domain CNN stream cannot carry a universality claim in either direction.

§8 Reporting commitment and stopping rule

- **Fixed N, no optional stopping.** All cells of all confirmatory seeds run. No hypothesis test is computed before the last cell is registered.
- Honest-reading clause: *"A paired test that does not reject is absence of evidence at this N. Absence of a practically relevant effect is claimed only through the ε-bounded states (BOUNDED-NO-GAIN, EQUIVALENT)."*
- All observed scalars are reported regardless of outcome, including `baseline`, `P_min`, `P_equ` cells, every secondary metric, the observed r and SD_D, and per-seed exclusions.

§9 Audit trail and deviations

- Cells registered via `harness/storage/run_registry.py` with profile keys `g4-nono/<arm>/<strategy>` (`<strategy>` = `na` for `baseline` and `P_min`) and R1 bit-stable run_ids.
- Milestones: `docs/milestones/g4-nono-cross-domain-<date>.{json,md}`.
- **Filing gate.** This draft is not filed until (a) the driver, `recombine_cross.py` and the aggregator exist at a pinned commit,
  (b) every item of the §5 fidelity checklist has a passing test,
  (c) the aggregator has been run on synthetic paired data covering every §2 state and every §7 row, and (d) the smoke results below (wall time, SD_D, $\hat{\tau}$ (descriptive), manipulation-check pass rate) and the N fixed by the §3 N-selection rule are written into the filed version. The lock commit is the commit that contains all four.
  Status 2026-10-24: (a) done: driver and `recombine_cross.py` (PR #44, three review rounds, APPROVE) and the aggregator (this PR), to be pinned by the lock commit; (b) done for items 1-4 and 6-8 on synthetic data; item 5 (new corpus) done on macM1, where the pinned corpus is built (it must be rebuilt and pass on the filing host);
  (c) done: `tests/unit/test_g4_nono_aggregator.py` reaches every §2 state and every §7 row end to end on synthetic paired records; (d) three smokes ran on 2026-10-24 and all are reported (§3): the first, at 3 epochs on CIFAR-100, hit the registered abort condition; the second, after the PI amendment (10 epochs, P_max recombine_full), passed and selected N = 95 on CIFAR-100; the third, on the new CIFAR-10 + Quick, Draw! corpus, passed and selected **N = 95**, the value to write into the filed version. **First smoke, 2026-10-24 (PI decision: pilot on macM1 now, campaign later).** Code at `28e18c5` (`DREAM0FKIKI_COMMIT_SHA`), macM1 (M1, 32 GB), CIFAR-100 from the pinned parquet mirror, one process at `nice 19`, run **concurrently with the N10-C campaign** (Metal near full load, swap 6.6-7.5 of 8 GB), so wall times may be inflated. 24 of 24 cells completed (`status: ok`, no cell-level exclusion, no non-finite metric); N10-C stayed alive throughout and the memory guard never triggered.
  - Wall time: driver-reported 7.7-16.3 s per cell (median 13.8 s; 289 s for 24 cells); 20-31 s per cell including process start and data loading; 613 s for the whole smoke.
  - Manipulation check (2-way task-1 accuracy, threshold 0.60): seed 900 0.609 (pass), 901 0.500 (fail), 902 0.526 (fail). **2 of 3 smoke seeds fail, which is the §4 abort condition**; the aggregator refuses the N selection (`n: null`).
  - Descriptive only (not used, because the selection is refused): SD_D of `WDPA(within) − WDPA(none)` = 0.0016 over the three seeds (below 0.015, so the rule would have given 95); $\hat{\tau}$ = 0.93 (CDPA cross/none), 0.99 (CDPA cross/within), 0.98 (WDPA within/none).
  - Final 20-way accuracy (`acc20`) is 0.05 in every `baseline` and `P_max` cell, DOA exactly 0.5, and CDPA / WDPA sit at 0.50-0.52. **Diagnosis (2026-10-24, short CPU reruns on macM1): not a pipeline bug.**

The 10 000 test images are balanced (500 per coarse class) and labelled with the coarse ids used by the 20-way head. After the stream, the shared head predicts almost only the last task's classes (seed 901: all 10 000 predictions on class 18; seed 900: 9 978 on class 19), the usual class-incremental collapse onto the most recent task. acc20 is then at most 0.1 × the last task's 2-way accuracy, and that pair (18 vs 19) stays at 0.50-0.503 2-way accuracy even at 10 epochs, so acc20 = 500 / 10 000 = 0.05. DOA is exactly 0.5 because every prediction falls in domain B (the last domain). At 10 epochs acc20 is 0.0503: more epochs do not move it. acc20 and DOA are therefore near-degenerate on this stream by construction (they are secondary endpoints). The manipulation-check failures are genuine under-fitting of a hard first pair (aquatic mammals vs fish, small CNN, plain SGD): test and train 2-way accuracy are equal (0.50-0.63), the cross-entropy ends near ln 2, and seed 901 predicts a single class after 3 epochs. 2-way task-1 accuracy by epochs (seed 900 / 901): 3 epochs 0.609 / 0.500, 8 epochs 0.601 / 0.613, 10 epochs 0.615 / 0.639; lr 0.05 at 3 epochs 0.586 / 0.500.

- Per the deviation envelope (b) below, this needs an amendment before any confirmatory cell runs (for example more epochs per G4-sexto §9.1); the choice is left to the PI. The smoke records are kept (`docs/milestones/g4-nono-smoke-records-2026-10/smoke1/`, ex macM1 `~/scratch/code/g4-pilot-out/`, with the runner log and the refused `n-selection.json`).
- Checklist item 5 was not run: no host has both pinned CIFAR-100 files (macM1 and Studio hold only the parquet mirror; electron-server holds neither). **Second smoke, 2026-10-24 (after the PI amendment).** Code at `c64bcee` (10 epochs, P_max recombine_full), same host, data, `nice` 19 and guard, a fresh output directory and registry, again **concurrently with N10-C** (swap 7.0-7.3 of 8 GB), so wall times may be inflated. 24 of 24 cells `ok`, no exclusion, no non-finite metric; N10-C stayed alive and the guard never triggered.
- Wall time: driver-reported 22-48 s per cell (median 25 s; 803 s for 24 cells); 40-61 s per cell including process start and data loading; 1 164 s for the whole smoke. Campaign estimate on macM1 at this load: about 48 s per cell, so ≈ 11.4 h for 855 cells (N = 95, 9 cells per seed; ≈ 10.2 h for the 760 confirmatory cells), ≈ 13 h for 972 cells (N = 108); with the driver time alone (median 25 s) about half of that.
- Manipulation check: 0.615 / 0.639 / 0.607 on seeds 900 / 901 / 902; all pass (margins over 0.60 are thin).
- **N selection: SD_D = 0.0022 < 0.015, so N = 95** (line 2). r̂ (descriptive): 0.999 (CDPA cross/none), 0.996 (CDPA cross/within), 0.992 (WDPA within/none).
- **P_equ and P_max now differ** in the `cross` and `within` cells (for example CDPA 0.5465 vs 0.5426 at seed 900 `cross`, 0.5203 vs 0.5246 at seed 901 `within`) and are bit-identical in the `none` cells, as designed (§5).
- CDPA / WDPA are 0.51-0.55; acc20 0.050-0.061 and DOA 0.5 in every cell (descriptive only, §6).
- Exploratory `P_max/unmixed` (run separately at `0b2b48e`, 3 cells): CDPA 0.5474 / 0.5257 / 0.5269, driver wall 24-46 s.
- Records: `docs/milestones/g4-nono-smoke-records-2026-10/smoke2/` (with `n-selection.json`) and `docs/milestones/g4-nono-smoke-records-2026-10/smoke2-unmixed/`. **Third smoke, 2026-10-24 (new corpus, PI decision; this is the smoke whose N is filed).** Code at `6f11823` (CIFAR-10 grayscale + Quick, Draw!, 10 epochs, P_max recombine_full), corpus `qd-cifar10-v1.npz` (SHA-256 `4f7fdf1c…`), same host, `nice` 19 and guard, fresh output directory and registry, again **concurrently with N10-C** (swap free ≥ 938 MB), so wall times may be inflated. 27 of 27 cells `ok` (24 confirmatory + 3 exploratory `unmixed`), no exclusion, no non-finite metric; N10-C stayed alive.
- Wall time: driver-reported 23-48 s per cell (median 44 s; 957 s for 27 cells); 1 094 s for the whole smoke, about 41 s per cell with start-up. Campaign estimate at this load: ≈ 10 h for 855 cells (N = 95), ≈ 11 h for 972 (N = 108).
- Manipulation check: 0.771 / 0.603 / 0.609 on seeds 900 / 901 / 902; all pass (901 and 902 only just).
- **N selection: SD_D = 0.0015 < 0.015, so N = 95** (line 2). r̂ (descriptive): 1.000 (CDPA cross/none), 1.000 (CDPA cross/within), 0.729 (WDPA within/none).
- CDPA rises to 0.62-0.82 (baseline 0.62-0.66, dream arms 0.67-0.82), WDPA 0.60-0.64: the cross-domain pairs are far easier to separate on this corpus (low-level cues other than colour remain, §5). acc20 0.09-0.12 and DOA ≈ 0.5 except one baseline cell (0.64), descriptive only.
- P_equ and P_max differ in `cross` / `within` and match in `none`, as designed. Exploratory `unmixed` CDPA 0.795 / 0.687 / 0.810.
- Checklist item 5 passes on this host for the new corpus.

  - Records: `docs/milestones/g4-nono-smoke-records-2026-10/smoke3/` (with `n-selection.json`).
  - **Re-check, 2026-10-25 (no rerun).** The aggregator changed after this smoke (`0705440`: exploratory cells split from the confirmatory ones before the abort, exclusion and completeness checks). `--select-n` at `d81652a` on the same 27 cell records (all `status: ok`, `commit_sha` `6f11823`) gives an output identical to the stored `n-selection.json` (24 confirmatory cells, no abort, line 2, **N = 95**). Recomputed by hand: the per-seed `WDPA(P_max/within) − WDPA(P_max/none)` are 0.00374, 0.00067 and 0.00191 (seeds 900 / 901 / 902), sd (ddof = 1) 0.00155. The smoke was not rerun and N is not re-selected.
- **Pre-filing smoke:** seeds 900-902, all 8 cells, on the filing hardware. It verifies the driver, the checklist tests and per-cell wall time, and records SD_D for the §3 N-selection rule (and $\hat{r}$, descriptive). Its numbers are reported but never pooled with confirmatory seeds.
- Known deviation envelopes:
  a. CIFAR-100 SHA-256 mismatch: abort and amend with the new pin.
  b. Manipulation-check failure on most seeds: abort per §4, raise epochs per G4-sexto §9.1 in the amendment.
  c. Per-cell wall > 60 s sustained: escalate to the PI before running further cells. Do not reduce N silently.
- Any other deviation needs an amendment commit **before** the affected cells run, or a post-hoc disclosure in Paper 2 that marks the affected hypotheses exploratory (per `docs/osf-preregistration-draft.md` §5).

§10 Open decisions for the PI (resolve before filing)

Everything above is the complete default protocol. The items below are genuine choices; changing one requires editing the named section before the lock commit.

Decided by the PI on 2026-10-23 and no longer open: ε = 0.01; N by the §3 N-selection rule (95 or 108; the N = 30 line was removed by the PI on 2026-10-24).

Within-domain equivalence margin (check, no change). The pre-registered within-domain margin already exists: H12-C is a paired TOST on WDPA with ε = 0.01 (§2), consistent with the G4 decision; a non-significant within-domain result is never read as "no effect" (§2, §8). No new margin is introduced. The domain × recombination interaction is read from H12-B (`cross` vs `within`), and retention versus cross-domain discrimination are reported as separate endpoints (§6).

1. ~~Domain corpus~~ — decided by the PI on 2026-10-24: CIFAR-10 (grayscale) + Quick, Draw! (§5).
2. ~~Un-mixed control arm~~ — decided by the PI on 2026-10-24: registered as an **exploratory** arm (§5), reported only.
3. **Hardware.** Partly decided by the PI on 2026-10-24: the smoke runs on macM1 now (done, see §9); the hardware for the confirmatory campaign is chosen after the N10-C campaign. The protocol does not change; R1 run_ids are per machine.
4. ~~Amendment after the smoke~~ — decided by the PI on 2026-10-24: epochs 3 → 10 (§5), acc20 and DOA descriptive only (§2, §6), and `P_max` runs recombine_full (§5). The smoke is rerun (§9).